

Electricity meter statistical types distinction

—— for three-phase meter from SolaX



1 Overview

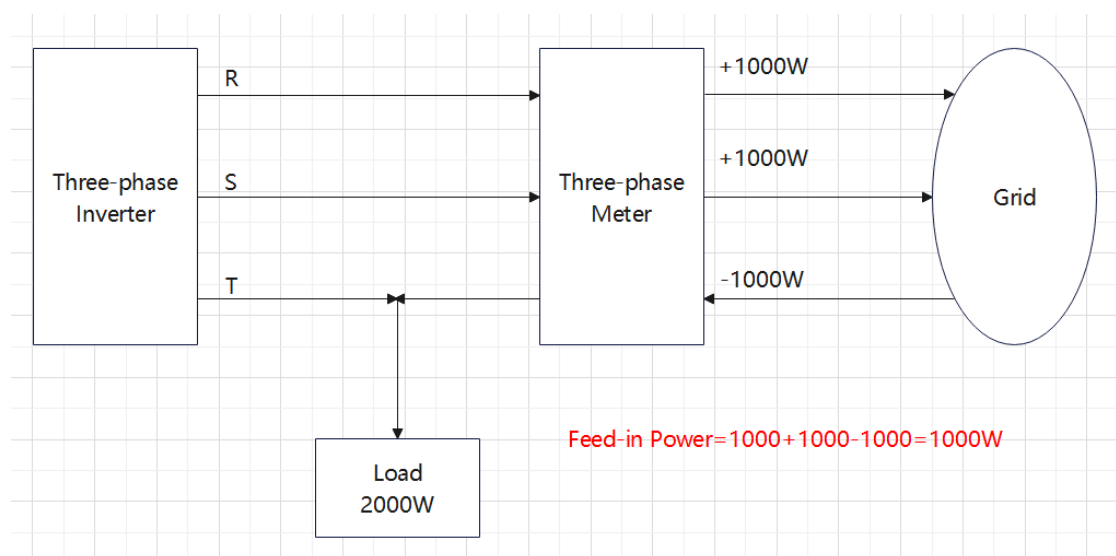
Electricity meter is important accessories in photovoltaic energy system. As a metering device, it supplies the feed-in power, feed-in energy and consumed energy of the system.

There are different methods to statistics these values **in three-phase system**. For the SolaX energy system, we divide statistical types into three: Phase Metering, Sum Metering and Net Metering.

2 Similarity

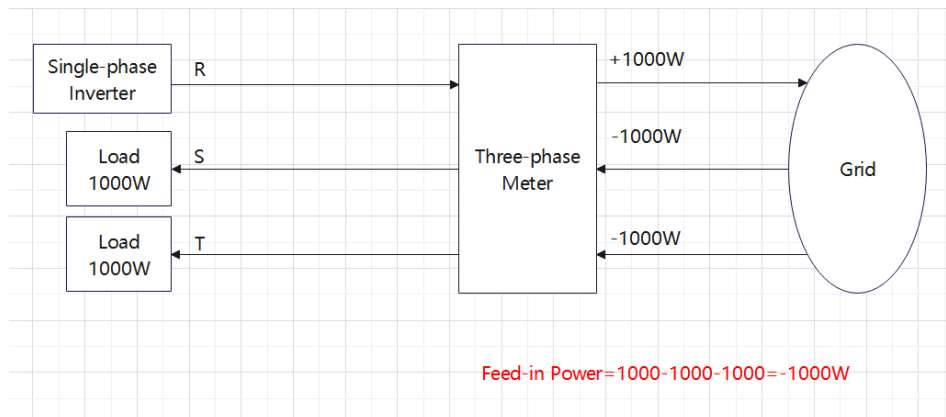
The statistics of feed-in power for these three types are same. They all use the three-phase algebraic sum as the total power display.

Example 1:



Two phase respectively sale 1000W electricity, one phase buys 1000W electricity. The total instantly feed-in power is sale 1000W as the algebraic sum of all three phase power.

Example 2:



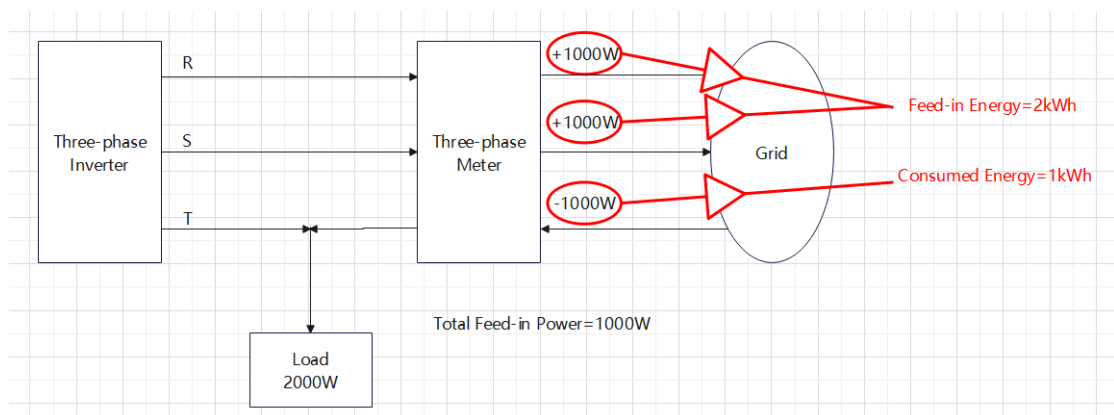
One phase sales 1000W electricity, two phase respectively buy 1000W electricity. The total instantly feed-in power is buy 1000W as the algebraic sum of all three phase power.

3 Distinction

3.1 Phase metering

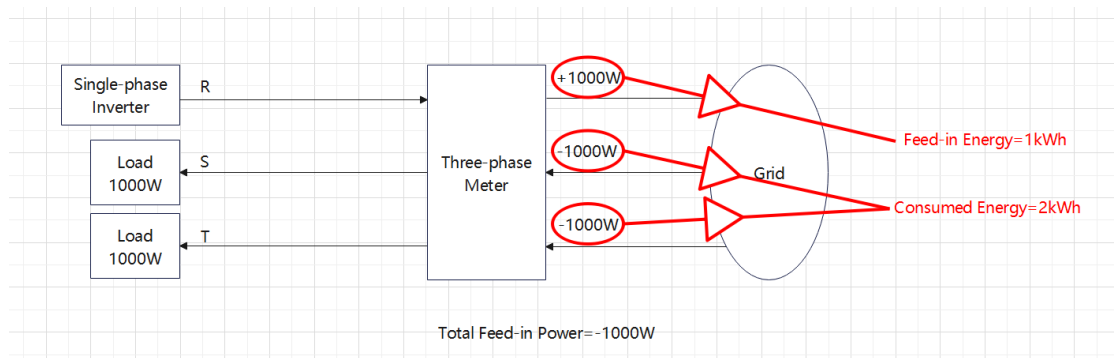
Statistical calculation type of feed-in energy and consumed energy for three-phase meter. The power of each phase is counted separately into feed-in energy or consumed energy.

For the example 1 above, if that status stable running for one hour:



There should be 2kWh feed-in energy and 1kWh consumed energy.

For the example 2 above, if that status stable running for one hour:

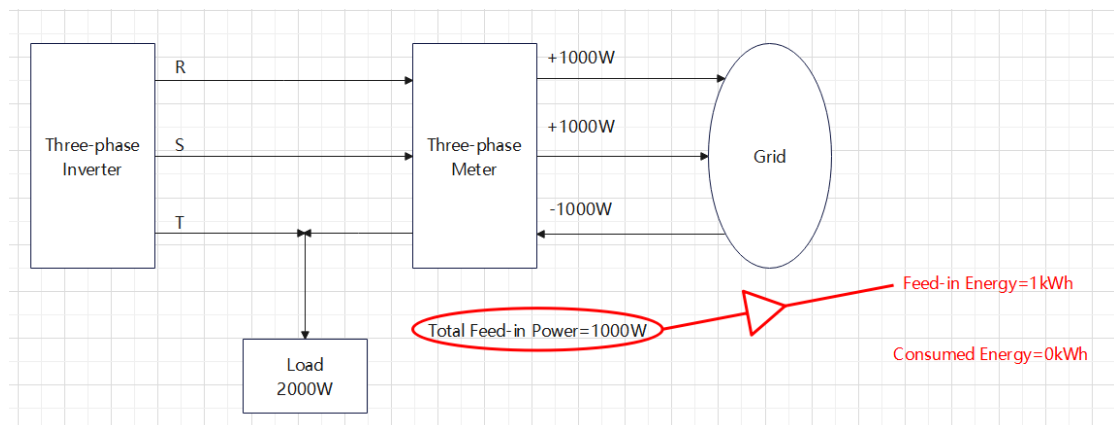


There should be 1kWh feed-in energy and 2kWh consumed energy.

3.2 Sum metering

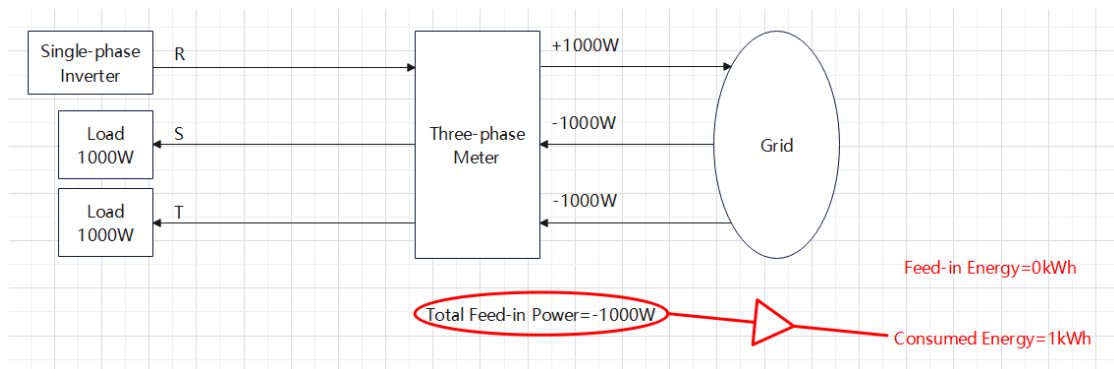
Statistical calculation method of feed-in energy and consumed energy for three-phase meter. The total feed-in power is counted into feed-in energy or consumed energy.

For the example 1 above, if that status stable running for one hour:



There should be 1kWh feed-in energy and no consumed energy.

For the example 2 above, if that status stable running for one hour:



There should be no feed-in energy and 1kWh consumed energy.

3.3 Net metering

Net metering is an electricity billing mechanism that allows consumers who generate some or all of their own electricity to use that electricity anytime, instead of when it is generated. Net metering policies can vary significantly by country and by state or province. Contact local energy suppliers to get more information. It will not count by the electricity meter from SolaX.

*Note: some meters from SolaX are labeled Net Metering, which actually stands for Sum Metering.

4 Identify and Setting

4.1 Distinguish the type of electricity meter from SolaX

If the following signs are present, the electricity meter is Sum Metering meter.

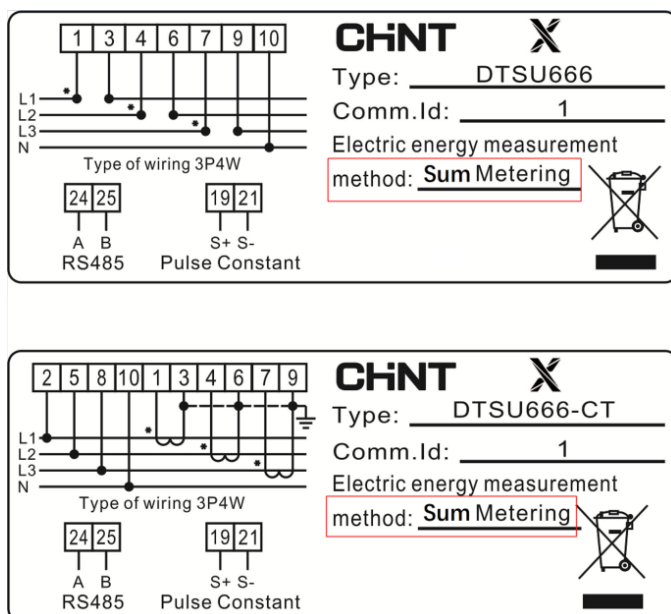
1) Sign 1 on the LCD screen.



2) Label on the meter.

There is Sum Metering shown on the label as method. It identifies the initial metering type of the meter.

*Note: some meters from SolaX are labeled Net Metering, which actually stands for Sum Metering.



4.2 Set the metering type on the LCD screen of electricity meter.

Check the E.SET item in the setting.



E.SET=0 means Phase Metering type;

E.SET=1 means Sum Metering type.